

LED Moving Message Display Simulator

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Presented here is a moving message display simulator that runs on a predefined six rows and 48 columns LED matrix. The LED dot-matrix-based moving message displays are popular devices which are used mainly as low-power, low-cost and high-productivity displays in various industrial applications, such as in steel factories and mills to display temperature, productivity index and to alert workers on site.

Railways, on the other hand, use these displays extensively on railway platforms for displaying train arrival information in real-time and also as coach display modules on the respective platforms; as digital clock consoles and as large LED colour screens to display the schedule of the trains and for video advertisements. You can also find them on the coaches to display information regarding the train's number, name, source and destination. Such displays are also used in banks for displaying foreign exchange rates, gold coin rates, fixed deposit rates and any other related information. These also find their use on suburban buses, BRTS (bus rapid transit system)/



Fig. 1: Screenshot of custom message

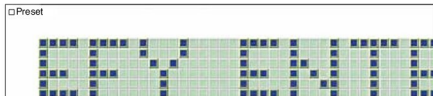


Fig. 2: Customised scrolling text message

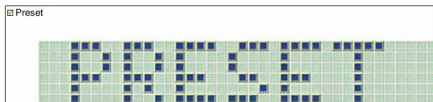


Fig. 3: Preset mode



MMTS (multi-modal transport system), metro trains, trams, etc.

The LED displays

The LED moving message display is nothing but an array of high-intensity LEDs arranged in rows and columns of specified lengths, which are controlled by the corresponding LED drivers from the circuit board. The LED drivers get decoded information from the decoder IC that encodes human-readable data into machine language and sends it to the display unit. These displays exploit the human persistence of vision phenomenon.

Persistence of vision. It is the phenomenon of the eye by which an afterimage persists for about one twenty-fifth of a second on the retina. This phenomenon is used mostly in cinematographic projectors that move the film in such a manner that the static

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

pictures produce a motion picture effect.

The program

The program 'LEDSMD.htm' is coded in JavaScript. When run, it displays a window with an array of simulated LEDs (similar to a dot-matrix-display). The intended message can be pre-defined in the code or customised. The array then starts scrolling out the intended message in the same style as seen in general displays. In actual physical LED displays, the LEDs do not move but appear to be moving due to the persistence of vision that gives an impression of scrolling.

The back-end JavaScript code has many functions that are defined so as to give the static object a moving effect, as seen in the LED displays. The program was tested on Windows XP with Mozilla Firefox version 22.0. It has two modes: custom message and preset message.

Custom message mode. In this mode, you can enter the text as per your requirement (refer Fig. 1). While testing we entered the text as 'EFY Enterprises Pvt Ltd, New Delhi.' When you press 'OK' button, you see the mentioned text scrolling from right to left of the screen (refer Fig. 2). The scrolling will continue till you exit the program. You can change the scrolling rate from the code defined at 'var scrollDelay=10'

Preset message mode. To display the preset message, just click the box against Preset option on top of the text, as shown in screenshot Fig. 3. The displayed text will be 'Preset message' scrolling from right to left. In this mode the 'Preset message' text is already entered in the code. The preset message can be directly defined in the code at the line setMessage('Text').

The code is well commented so as to make it easy to go through and understand without much difficulty. ●

The author is a B.Tech in electronics and communication who likes coding in Javascript. He has a keen interest in space technology (satellite communication) and mobile communication